INTRODUCTION TO DEEP LEARNING

Course number: 00240332

Lecture 11: Sentence processing

Xiaolin Hu (胡晓林)

Dept. of Computer Science and Technology

Tsinghua University

Last lecture review

I. Introduction to NLP

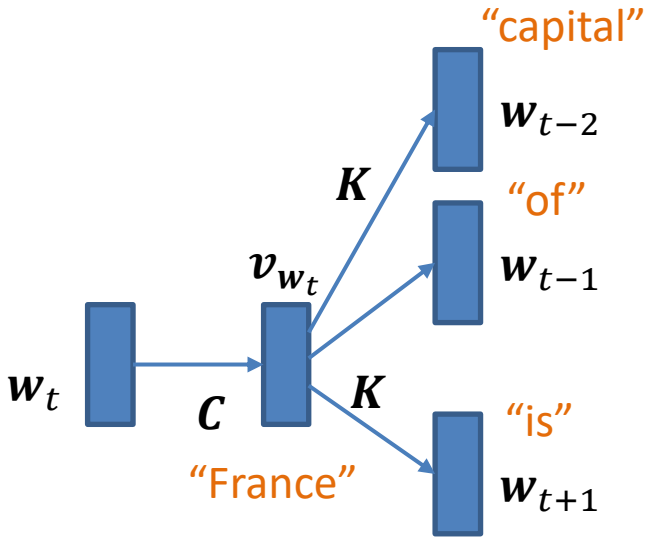
- Tagging and parsing
- Text generation
- Machine translation
- Text/document classification
- Question answering
- Sentiment analysis

II. Word representation

- Co-occurrence matrix
- Dense vector

counts	I	like	deep	learning	NLP	enjoy	flying
I	0	2	0	0	0	1	0
like	2	0	1	0	1	0	0
deep	0	1	0	1	0	0	0
learning	0	0	1	0	0	0	0
NLP	0	1	0	0	0	0	0
enjoy	1	0	0	0	0	0	1
flying	0	0	0	0	0	1	0

III. Word2vec



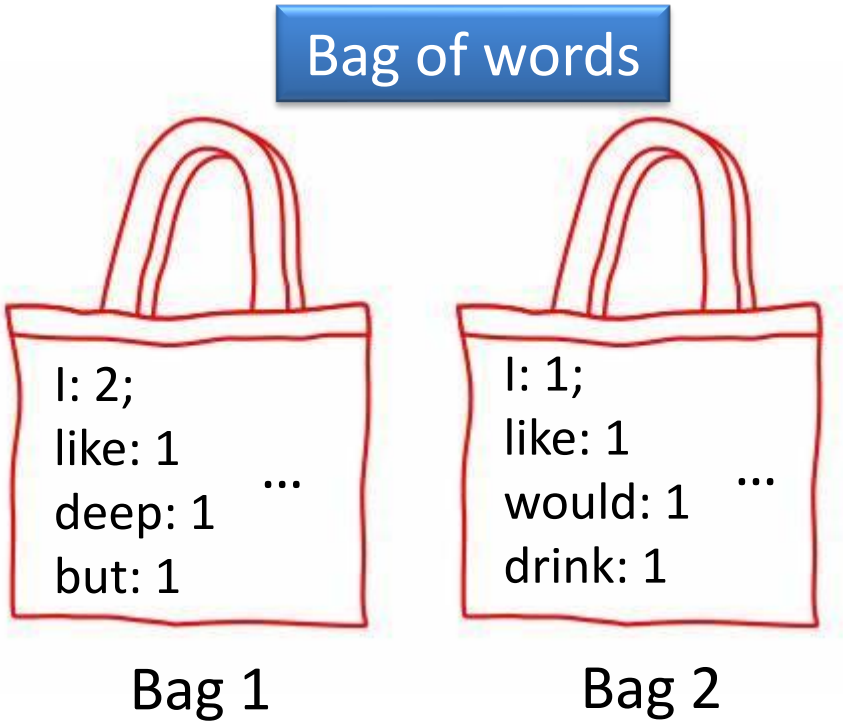
Paper sharing

- Every course project group prepares a 5-minute presentation with slides for one of the following papers
  - Liang, Hu, “Recurrent convolutional neural network for object recognition,” CVPR 2015
  - Bengio et al., “A neural probabilistic language model,” Journal of Machine Learning Research, 2003

Without word embedding, how do people represent sentences/documents?

"I like deep learning, but I do not have much time for doing homework."

"I would like to drink a cup of coffee."



Disadvantage: lack of order of the words

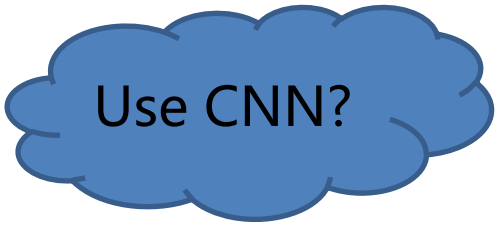
"white blood cells destroying an infection"

"an infection destroying white blood cells"

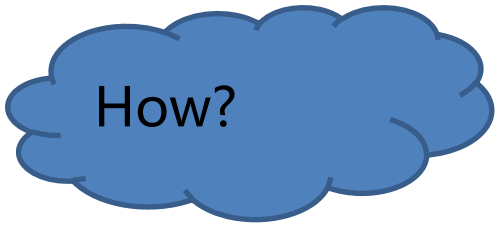
With word embedding, how do we classify sentences/documents?

"I like deep learning, but I do not have much time for doing homework."

"I would like to drink a cup of coffee."





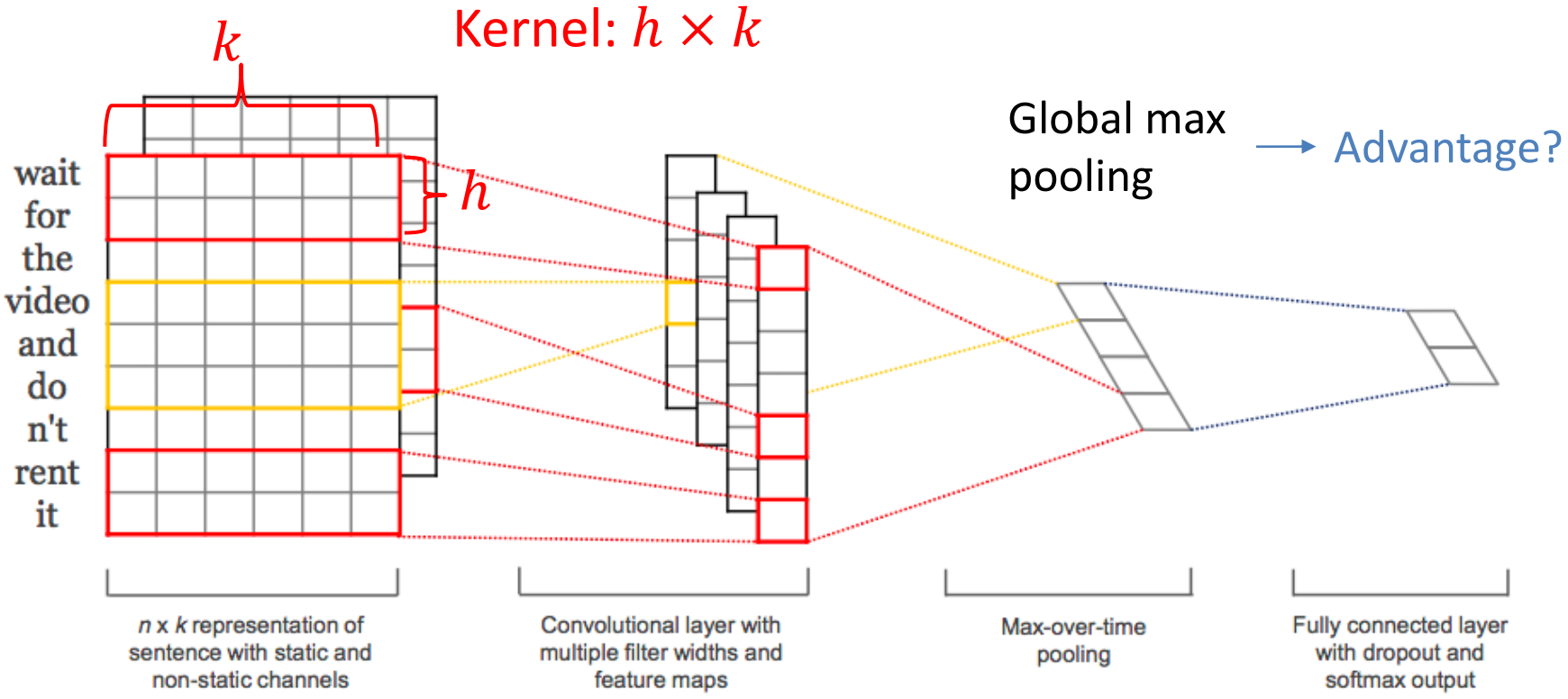


Outline

I. Convolutional neural networks
II. Recurrent neural networks
III. Summary

Text classification using CNNs

Kim, 2014



Each word has a k-dim real vector

Each vector corresponds to a different kernel

Simple idea, great influence



Yoon Kim

FOLLOW

Assistant Professor, MIT
Verified email at mit.edu - Homepage

Machine Learning Natural Language Processing Deep Learning

TITLE	CITED BY	YEAR
Convolutional Neural Networks for Sentence Classification Y Kim EMNLP 2014	20556 *	2014
OpenNMT: Open-Source Toolkit for Neural Machine Translation G Klein, Y Kim, Y Deng, J Senellart, AM Rush ACL 2017 (System Demonstrations)	2481	2017
Character-Aware Neural Language Models Y Kim, Y Jernite, D Sontag, AM Rush AAAI 2016	2206	2015
Sequence-Level Knowledge Distillation Y Kim, AM Rush EMNLP 2016	1136	2016

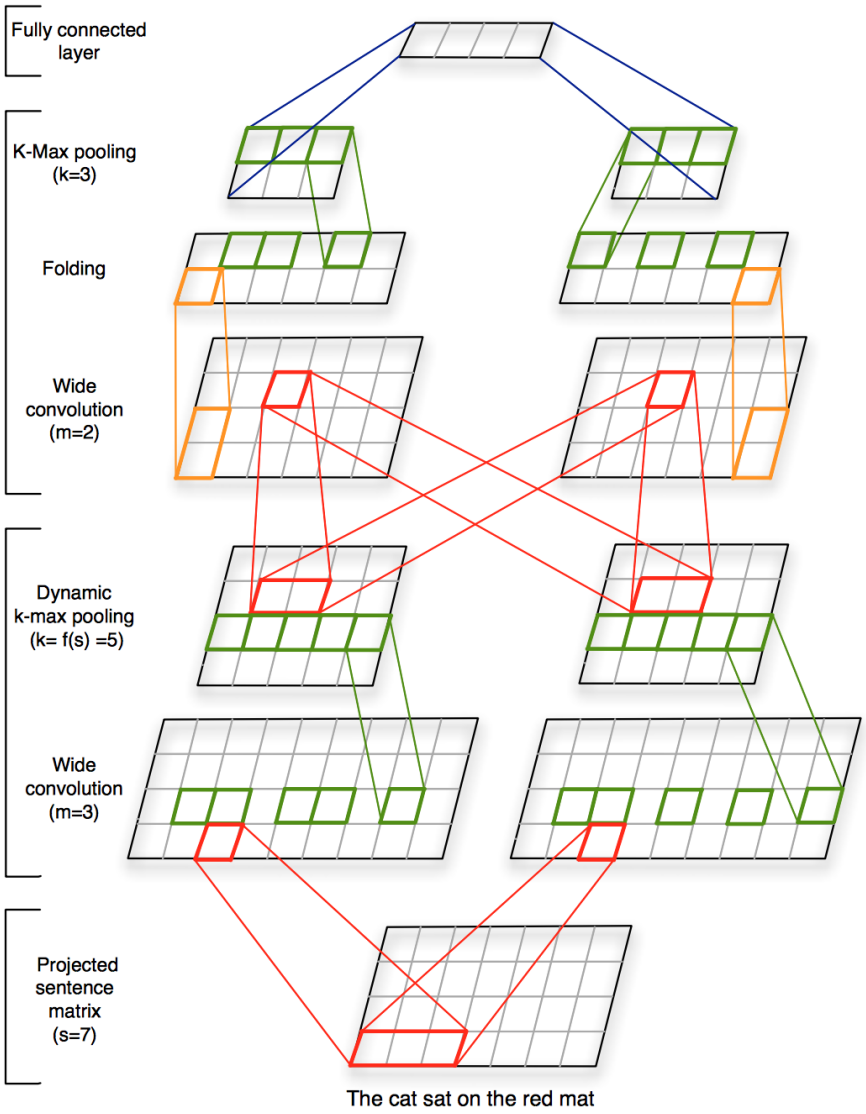
From Google Scholar, 2024/11/25

Any problem with this model?

- It's not deep enough
  - One conv layer and one pooling layer
- Features are not diverse enough
  - Each convolutional kernel results in a 1D feature map, i.e., a feature vector
  - The global max pooling on one feature vector, i.e., max pooling over all time, results in a scalar

What's the major difficulty for constructing deeper models?

A deeper model

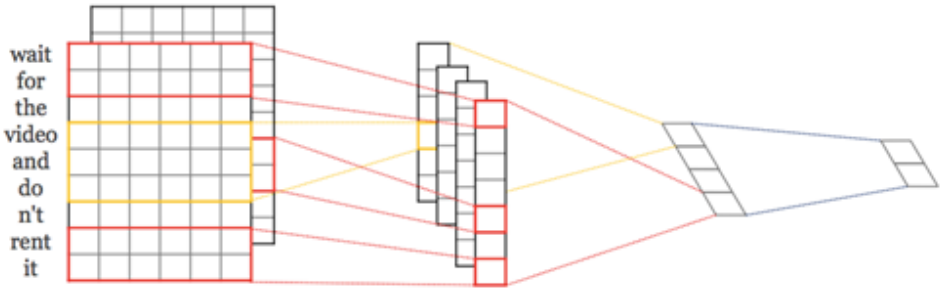


Kalchbrenner et al., 2014

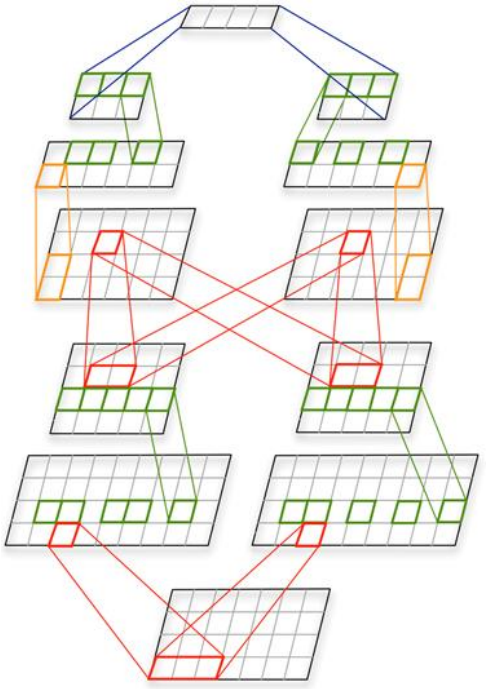
- 1-D convolution along the time axis
- K-max pooling over time
- Folding: elementwise summation of two rows
  - Like a special average pooling

Summary of Part I

Shallow CNN



Deep CNN



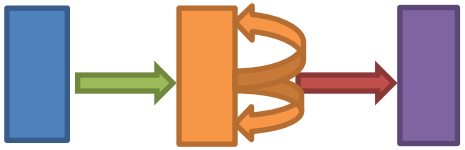
Outline

I. Convolutional neural networks
II. Recurrent neural networks
III. Self-attention
IV. Summary

Seq2seq is from Stanford CS224N

Text classification using RNNs

Unfold for the Elman network

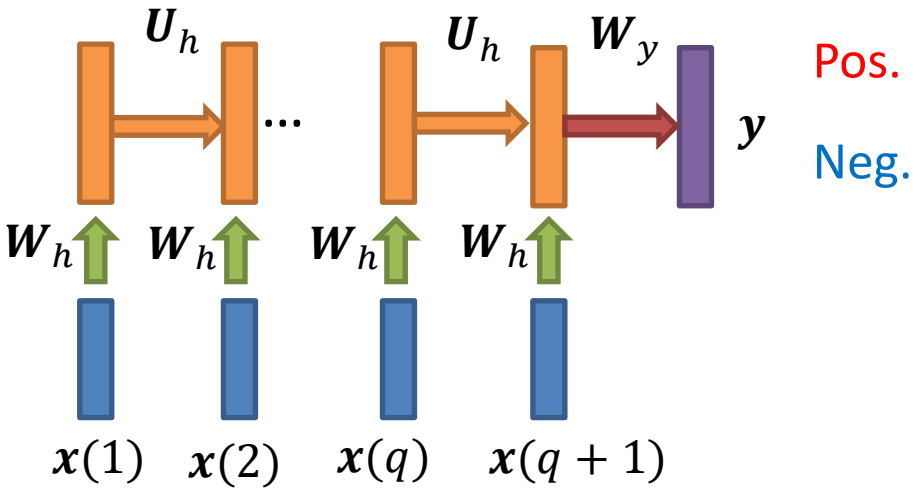


Input $\boldsymbol{x}$ Hidden $\boldsymbol{h}$ Output $\boldsymbol{y}$

$$\boldsymbol{h}(t) = \sigma_h(\boldsymbol{W}_h\boldsymbol{x}(t) + \boldsymbol{U}_h\boldsymbol{h}(t-1) + \boldsymbol{b}_h)$$

$$\boldsymbol{y}(t) = \sigma_y(\boldsymbol{W}_y\boldsymbol{h}(t) + \boldsymbol{b}_y)$$

- $\boldsymbol{x}$ is time-varying
- Label $\boldsymbol{r}$ is only present at the last step

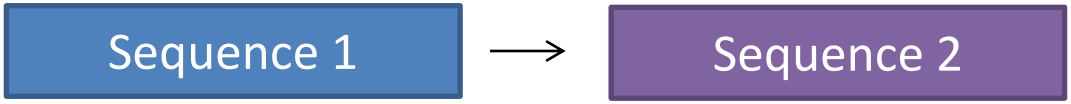


- LSTM or GRU can be used
- Bidirectional RNN can be used
- See homework

Deep learning is so cool!

His attitude is annoying!

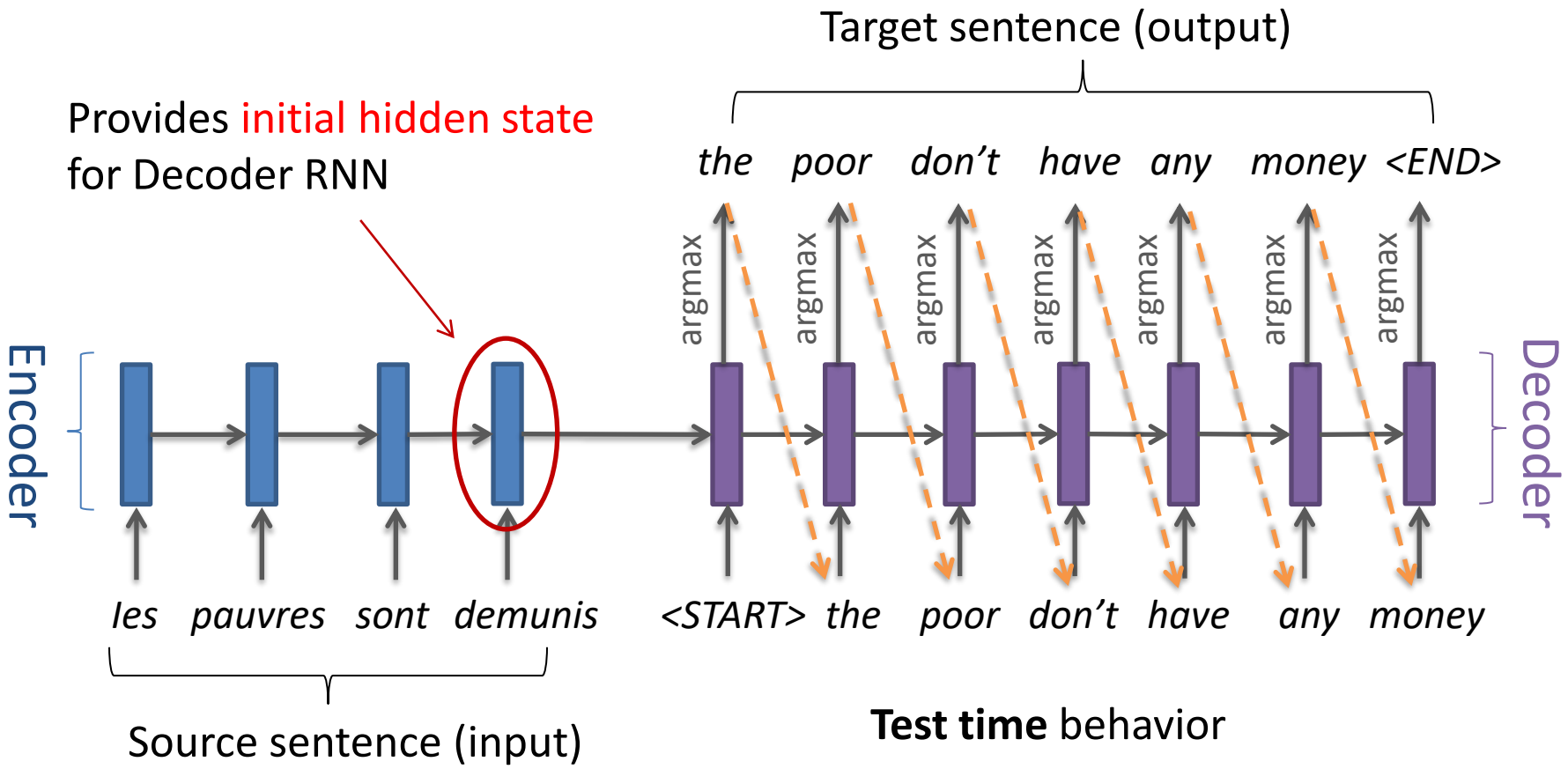
Sequence-to-sequence learning



- Usually it involves two RNNs: encoder and decoder
- Many NLP tasks can be phrased as sequence-to-sequence:
  - Machine translation (French → English)
  - Summarization (long text → short text)
  - Dialogue (previous utterances → next utterance)
  - Parsing (input text → output parse as sequence)
  - Code generation (natural language → Python code)
  - Melody generation (one musical phrase → next phrase)
  - Speech recognition (sound → text)

Neural machine translation (NMT)

Sutskever et al., 2015



- Encoder RNN produces an encoding of the source sentence
- Decoder RNN generates target sentence conditioned on encoding

单选题 1分

设置

If you use the CE loss to train the model, which of the following should NOT be used to represent words?

A One-hot representation input to the encoder

B Pretrained word embeddings input to the encoder

C One-hot representation output by the decoder

D Pretrained word embeddings output by the decoder

提交

Encoder and decoder

- For the encoder, one can use either pretrained word embedding, e.g., word2vec, or one-hot representation
- For the decoder, the one-hot representation is used
  - We need a prob for each reference word to define the objective function
  - Softmax function is usually used as the output
  - What if the dictionary is very large?
- The encoder RNN and decoder RNN are often different, and deep RNNs can be used

单选题 1分

Setting

Can the dictionaries for the encoder and the decoder be different?

A Yes

B No

Submit

多选题 1分

Setting

In which tasks, can the dictionaries for the encoder and the decoder be the same?

A Machine translation

B Text summarization

C Story generation

D Dialog

Submit

Conditional Language Model

- The sequence-to-sequence model is an example of a Conditional Language Model
  - Language Model because the decoder is predicting the next word of the target sentence *y*
  - Conditional because its predictions are *also* conditioned on the source sentence *x*
- NMT directly calculates $P(\boldsymbol{Y}|\boldsymbol{X})$, where $\boldsymbol{Y} = [\boldsymbol{y}_1, \dots, \boldsymbol{y}_T]$, $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_{T'}]$

$$P(\boldsymbol{Y}|\boldsymbol{X}) = P(\boldsymbol{y}_1|\boldsymbol{X})P(\boldsymbol{y}_2|\boldsymbol{y}_1, \boldsymbol{X})P(\boldsymbol{y}_3|\boldsymbol{y}_1, \boldsymbol{y}_2, \boldsymbol{X}) \cdots \underbrace{P(\boldsymbol{y}_T|\boldsymbol{y}_1, \dots, \boldsymbol{y}_{T-1}, \boldsymbol{X})}$$

Probability of next target word, given target words so far and source sentence $\boldsymbol{X}$

- How to train an NMT?
  - You need a big parallel corpus…

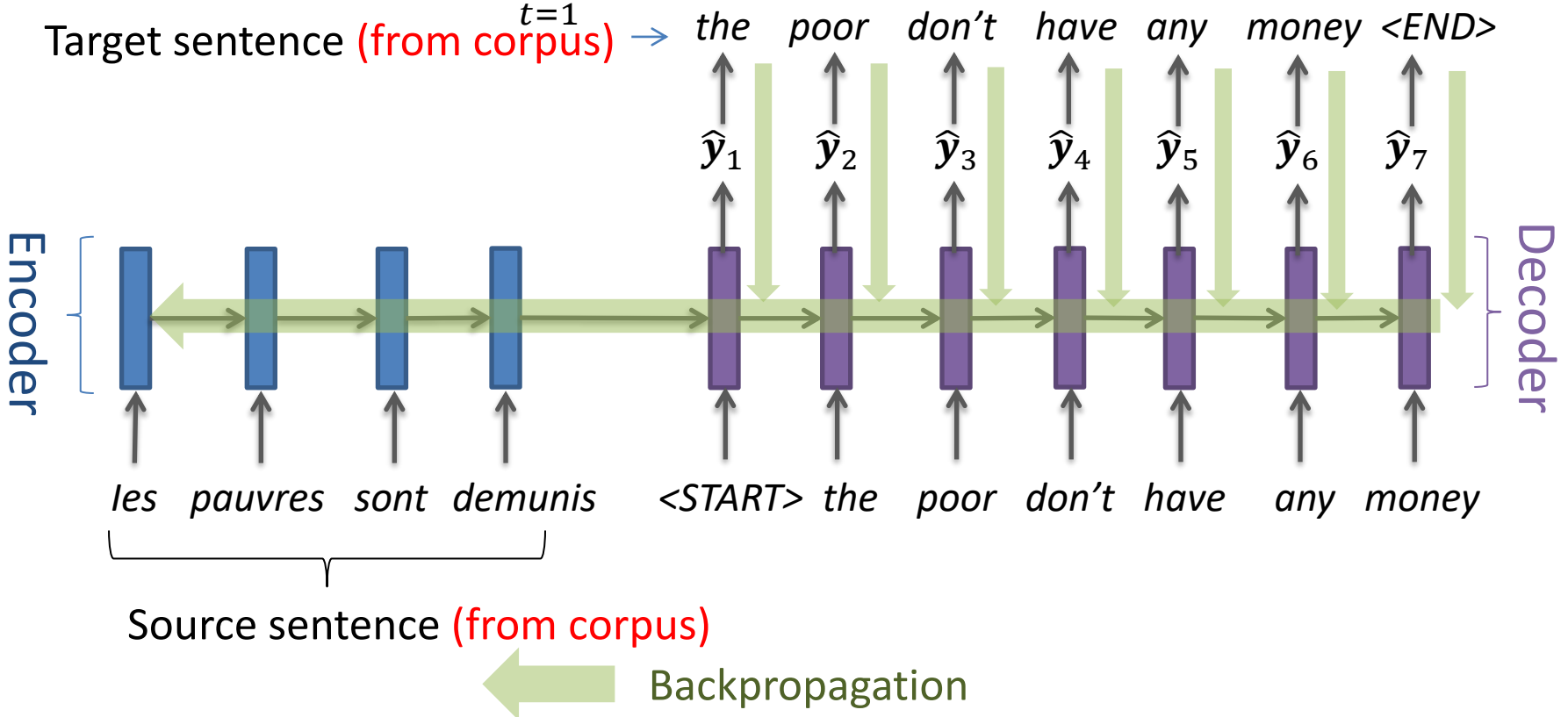
Training an NMT system

= negative log prob of "the"

= negative log prob of "have"

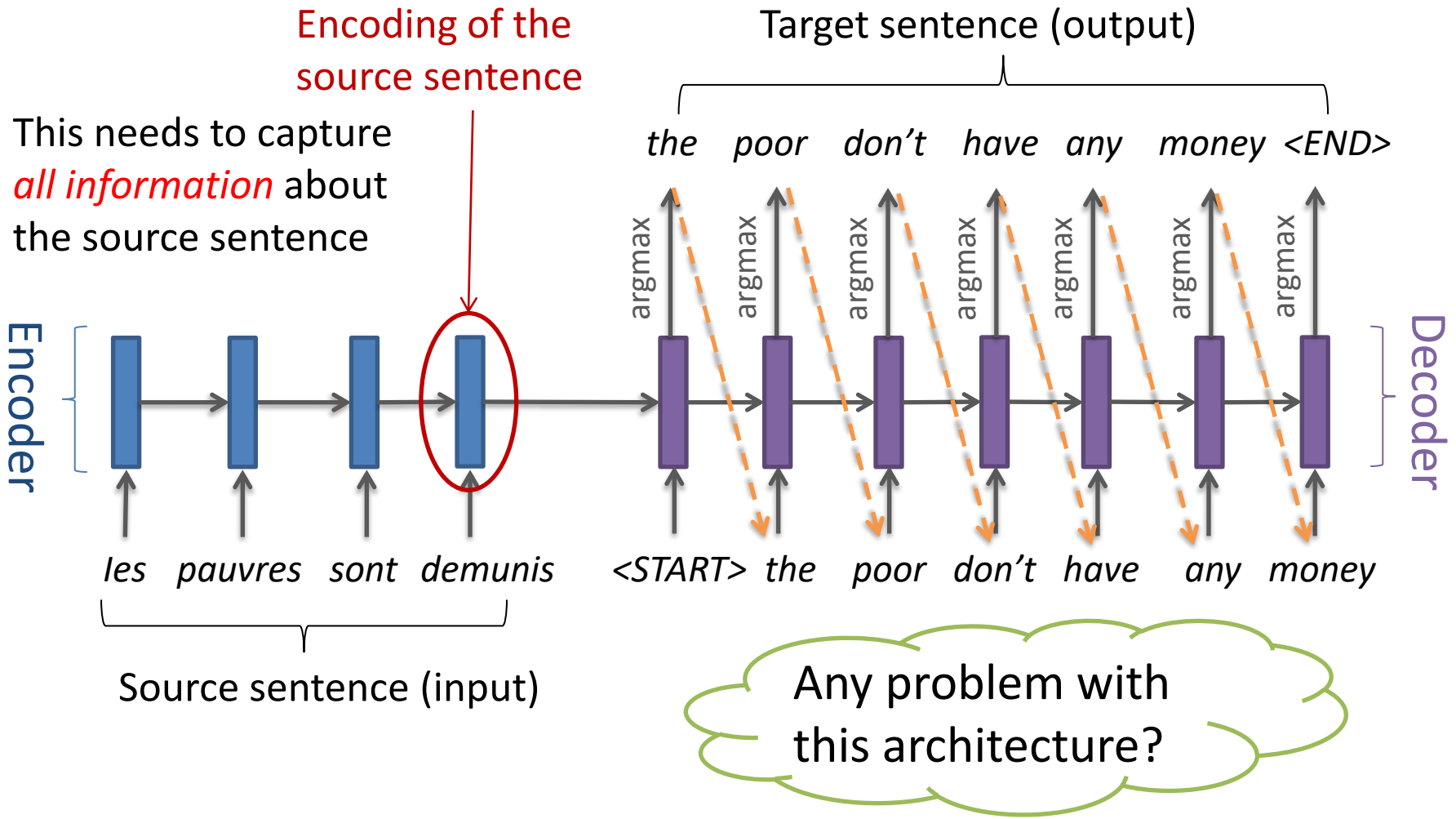
= negative log prob of <END>

$$J = \frac{1}{T}\sum_{t=1}^{T} J_t = J_1 + J_2 + J_3 + J_4 + J_5 + J_6 + J_7$$



Seq2seq is optimized as a **single system**

The bottleneck problem



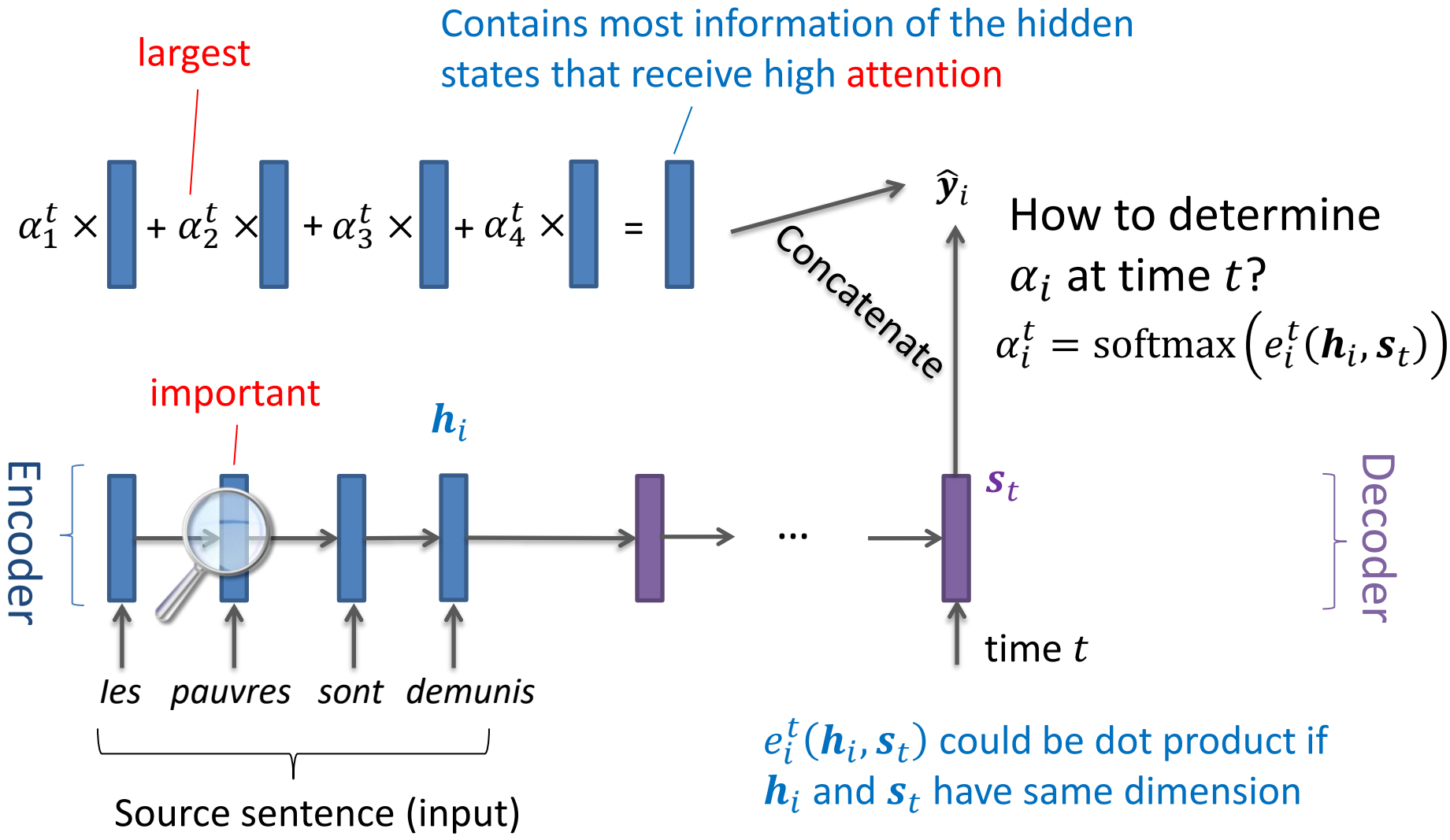
Attention

- Attention provides a solution to the bottleneck problem.
- Core idea: on each step of the decoder, *focus on a particular part* of the source sequence



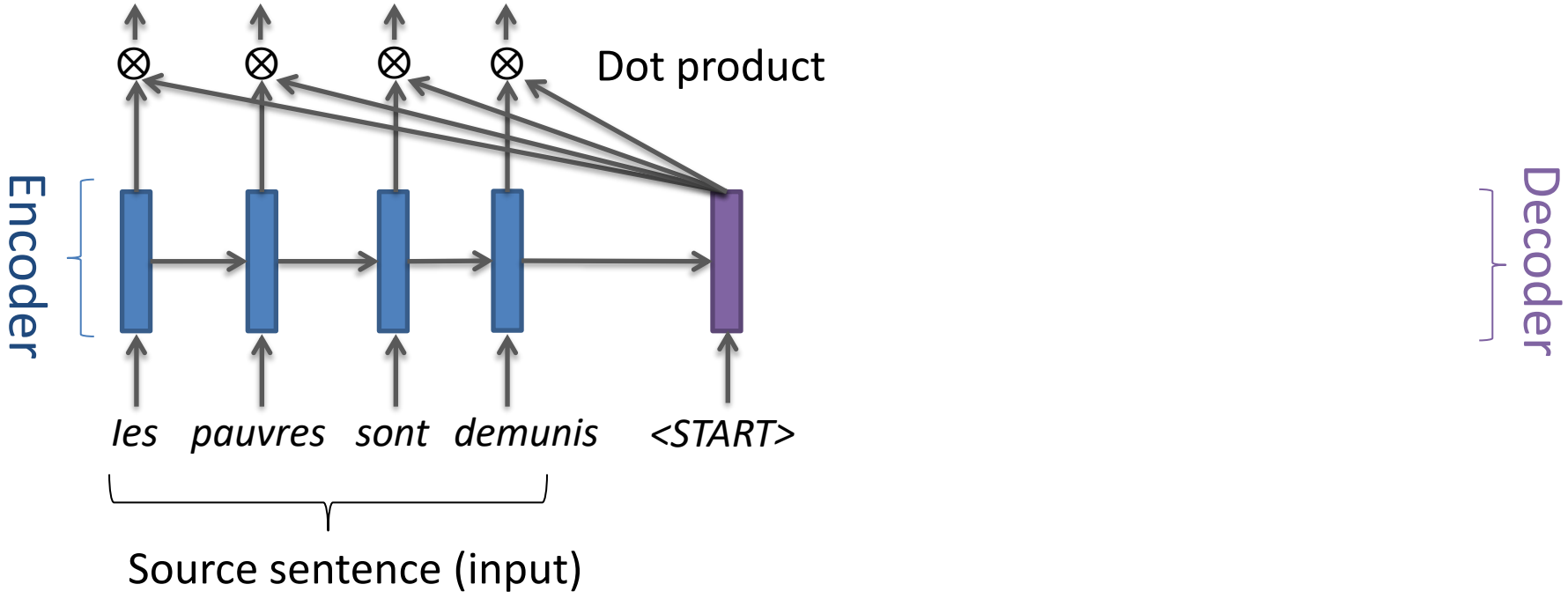
- There are many forms of attention
- I will show a simple example

What do we want?



Seq2seq with attention

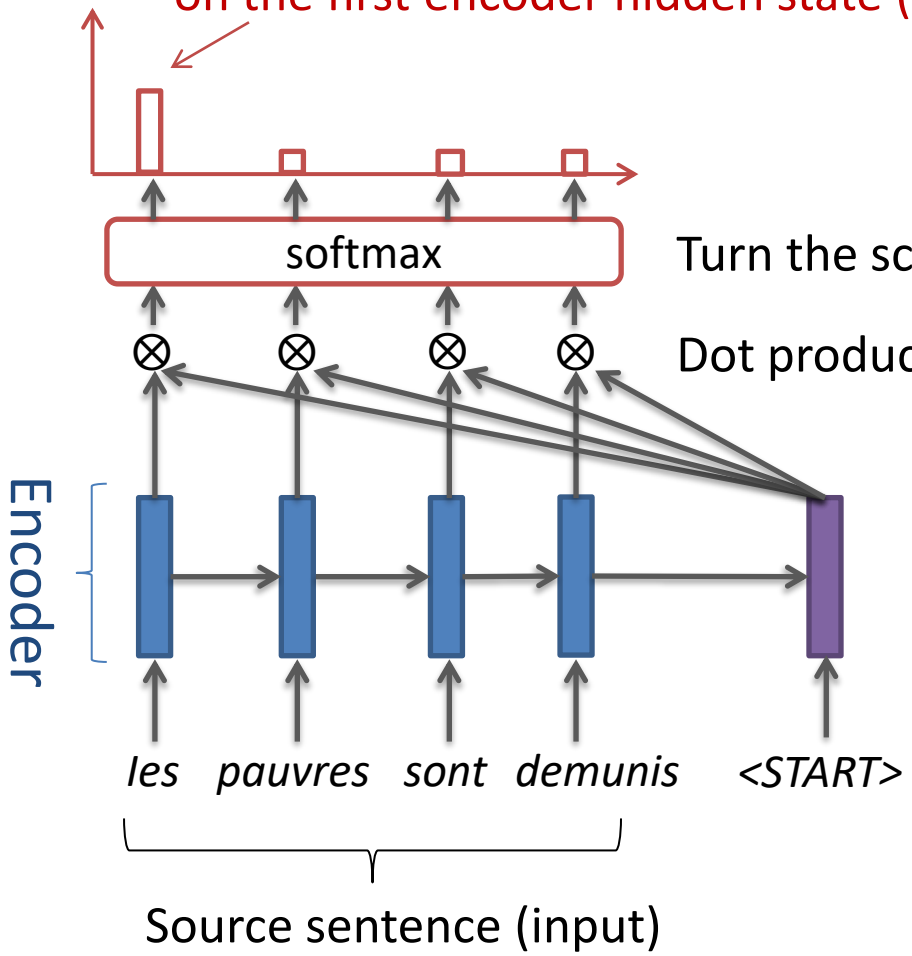
Bahdanau et al., 2015



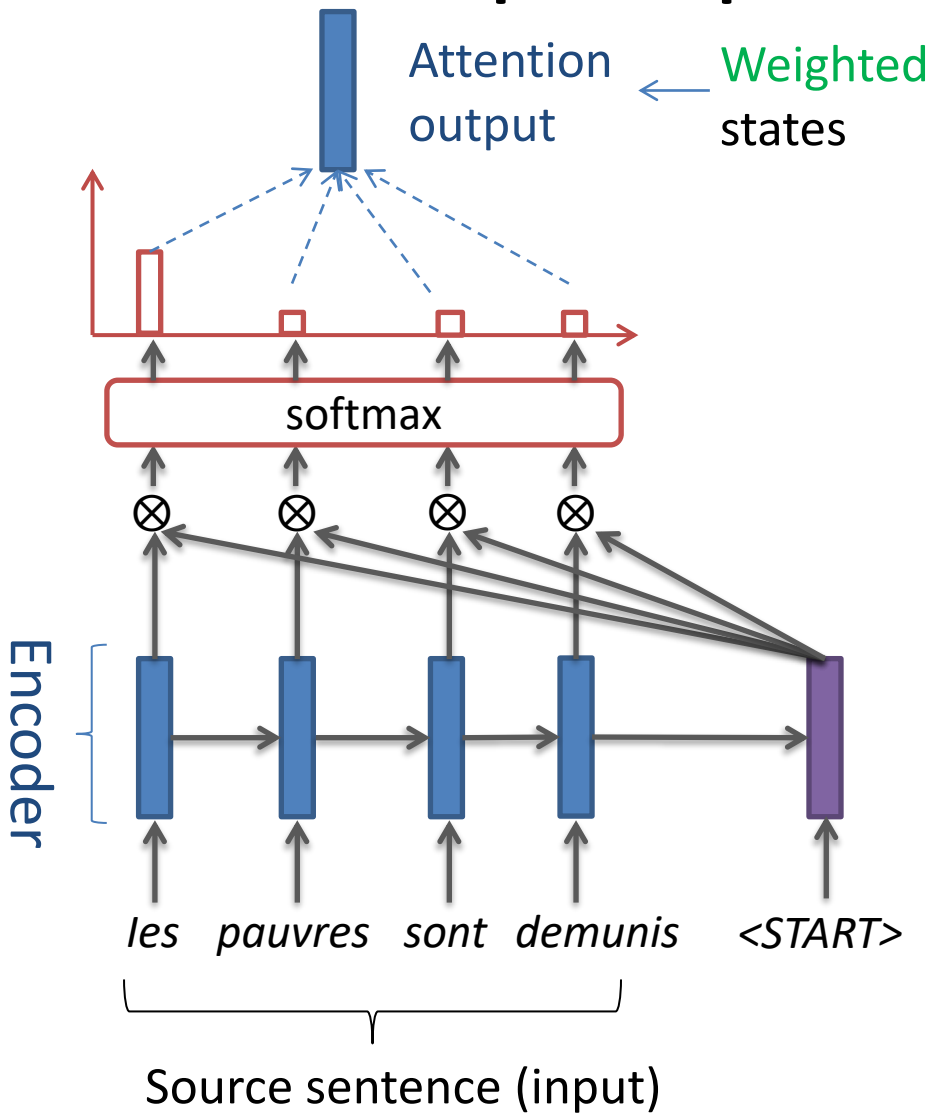
Seq2seq with attention

Bahdanau et al., 2015

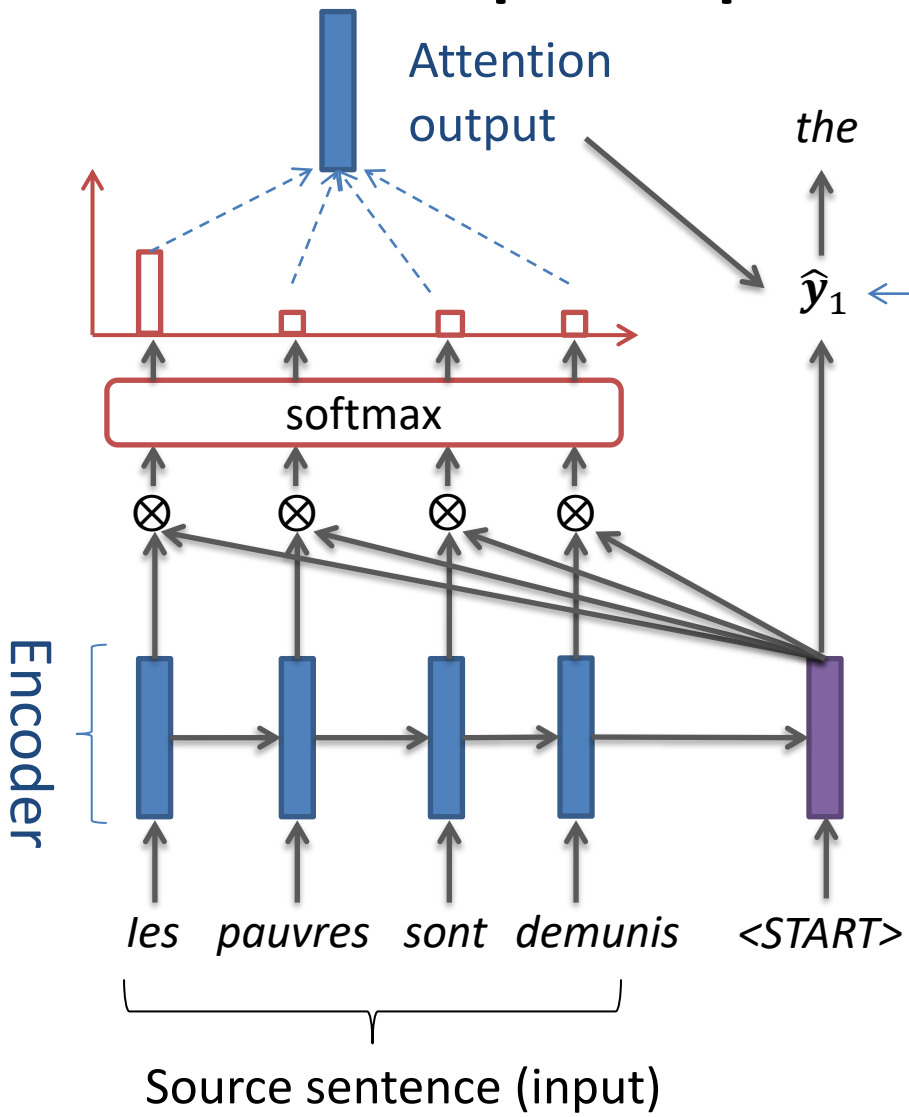
On this decoder time step, we're mostly focusing on the first encoder hidden state (*"les"*)



Seq2seq with attention

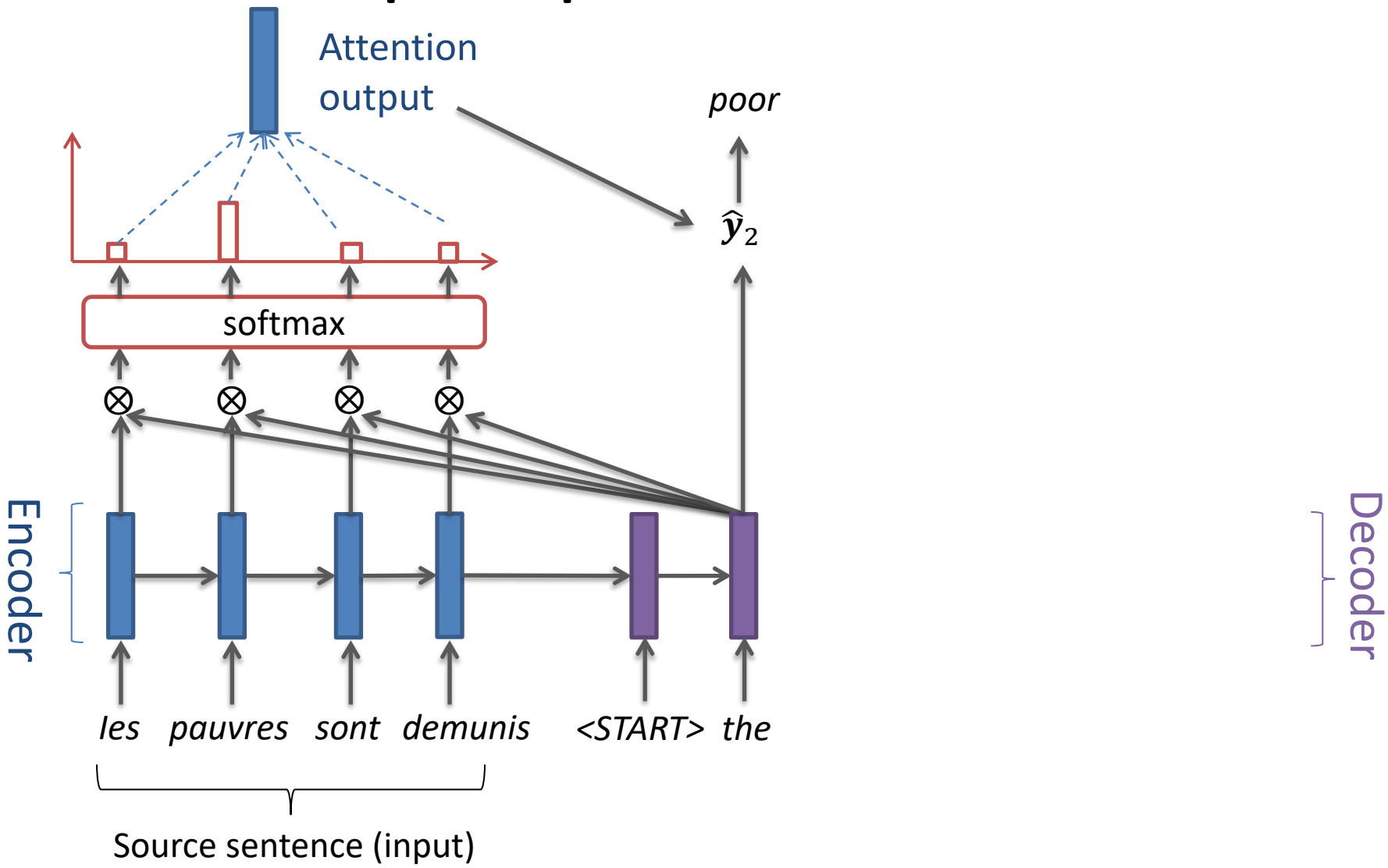


Seq2seq with attention

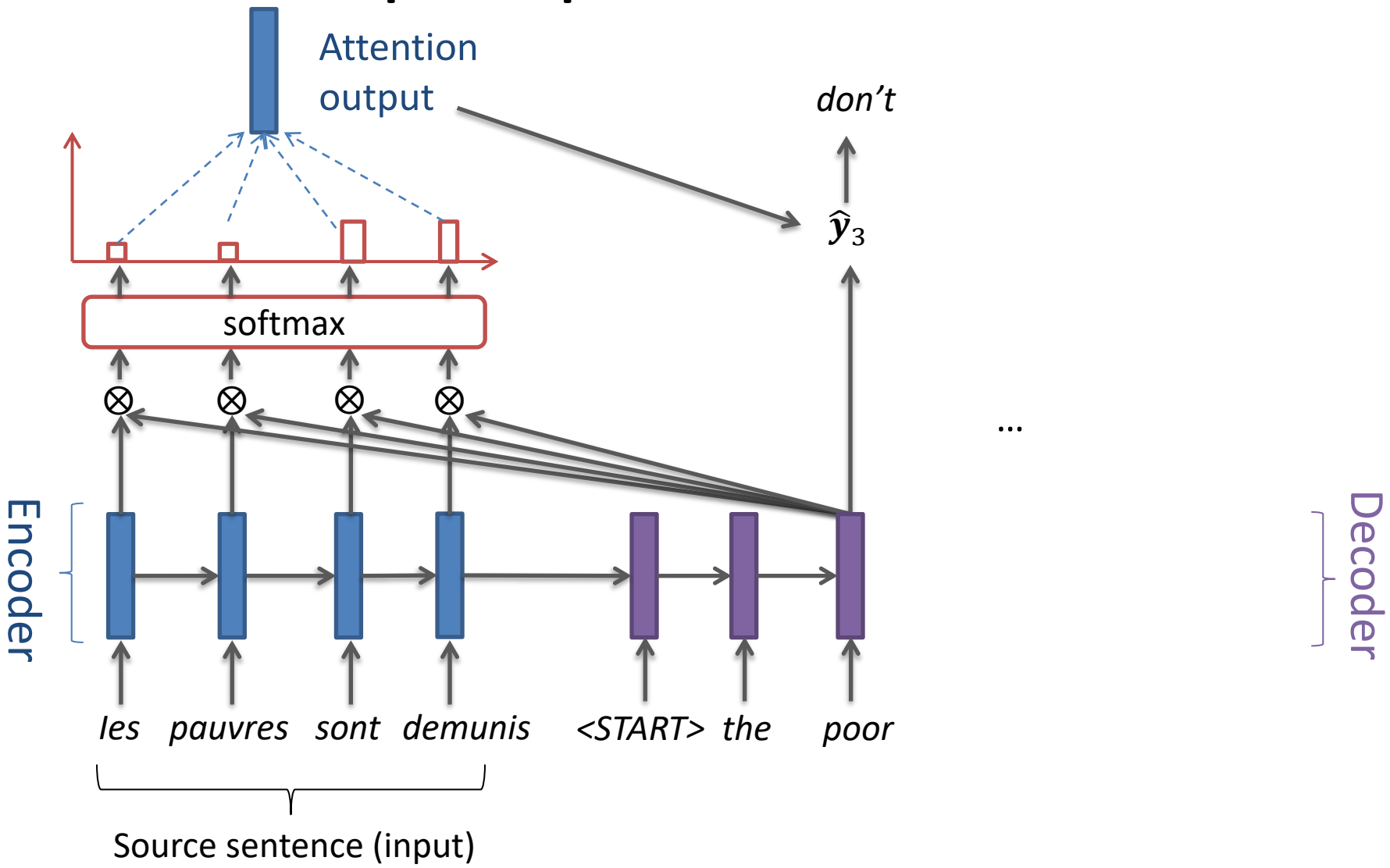


Concatenate attention output with decoder hidden state, then use the result to compute $\hat{y}_1$ as before

Seq2seq with attention



Seq2seq with attention



Formulation

The encoder's hidden states: $\boldsymbol{h}_1, \ldots, \boldsymbol{h}_N \in R^d$
At time step t, the decoder's hidden state $\boldsymbol{s}_t \in R^d$

- We get the attention scores $\boldsymbol{e}^t$ for this step:

$$\boldsymbol{e}^t = [\boldsymbol{s}_t^\top \boldsymbol{h}_1, \ldots, \boldsymbol{s}_t^\top \boldsymbol{h}_N] \in R^N$$

- Take softmax to get the attention distribution for this step

$$\boldsymbol{\alpha}^t = \text{softmax}(\boldsymbol{e}^t) \in R^N$$

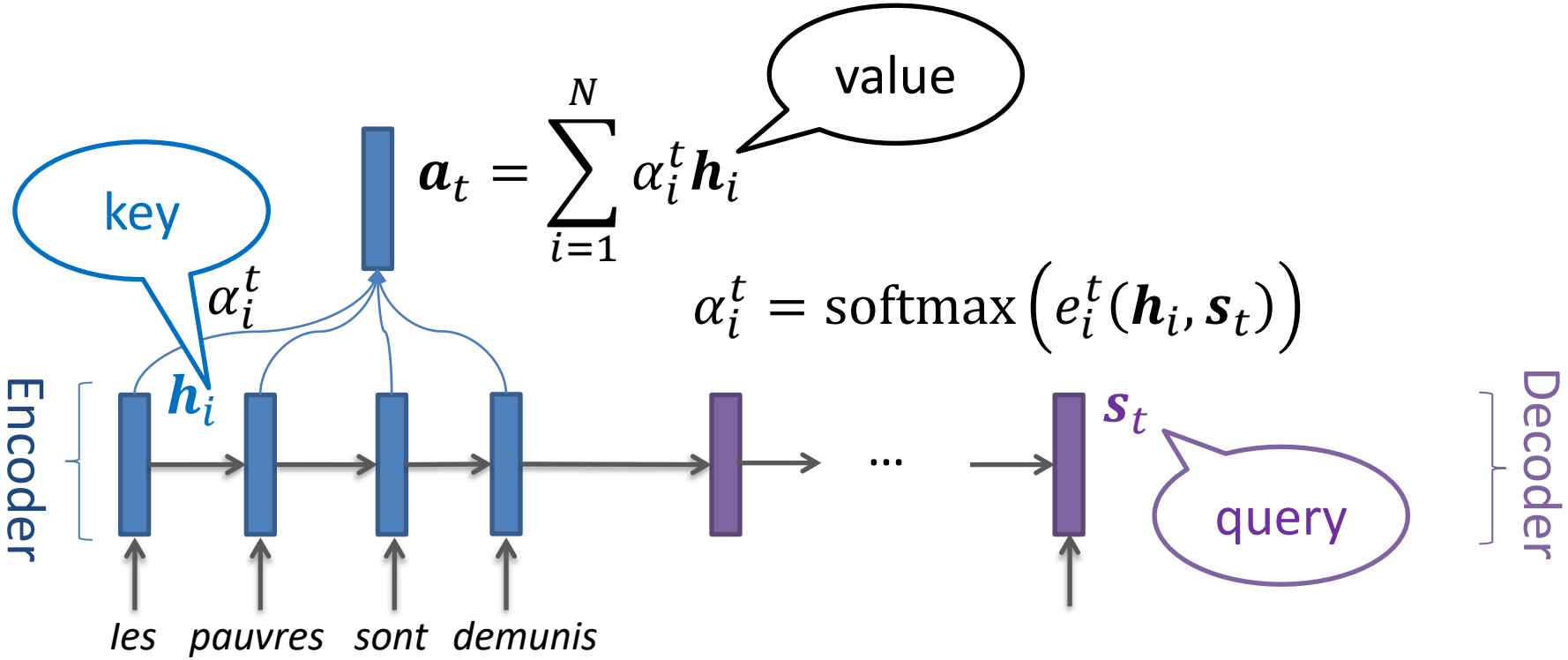
- Calculate the attention output

$$\boldsymbol{a}_t = \sum_{i=1}^{N} \alpha_i^t \boldsymbol{h}_i \in R^d$$

- Finally, concatenate the attention output with the decoder hidden state and proceed as in the non-attention seq2seq model

$$[\boldsymbol{a}_t; \boldsymbol{s}_t] \in R^{2d}$$

Cross attention

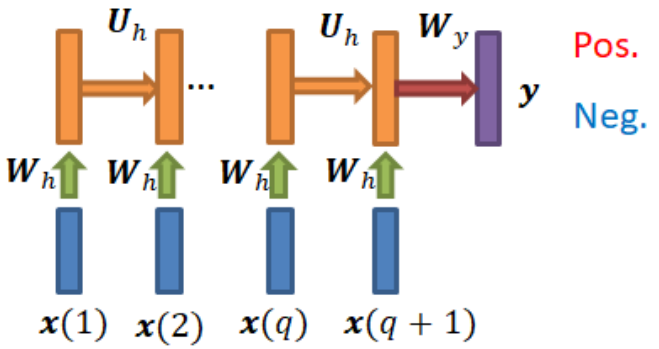


$$\boldsymbol{a}_t = \sum_{i=1}^{N} \mathrm{softmax}\left(e_i^t(\boldsymbol{h}_i, \boldsymbol{s}_t)\right) \boldsymbol{h}_i$$

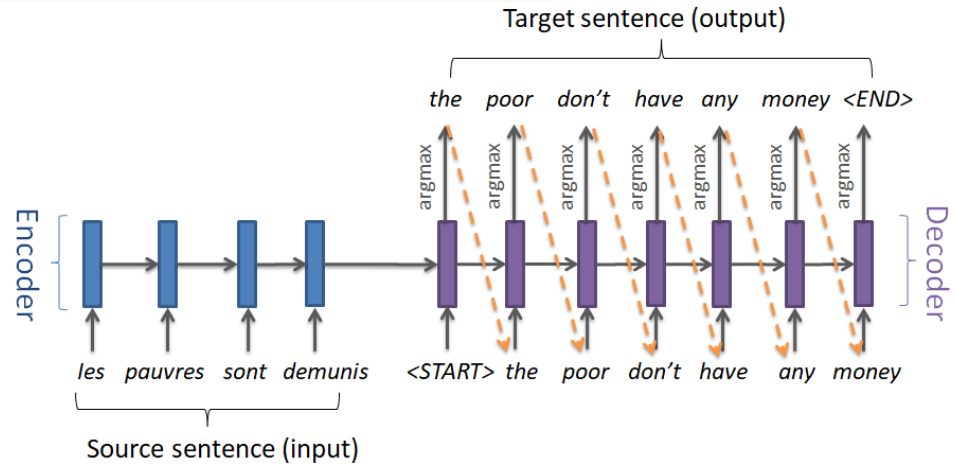
key query value

Summary of Part II

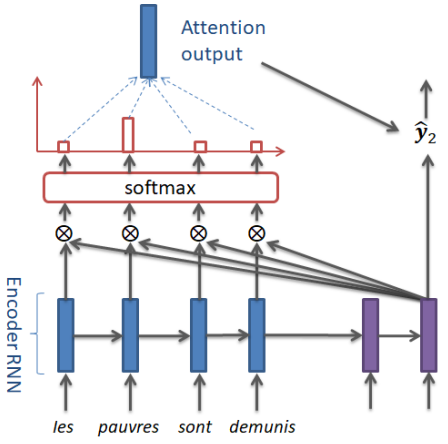
RNN for sentence classification



Seq2seq for NMT



Seq2seq w/ attention



Outline

I. Convolutional neural networks
II. Recurrent neural networks
III. Summary

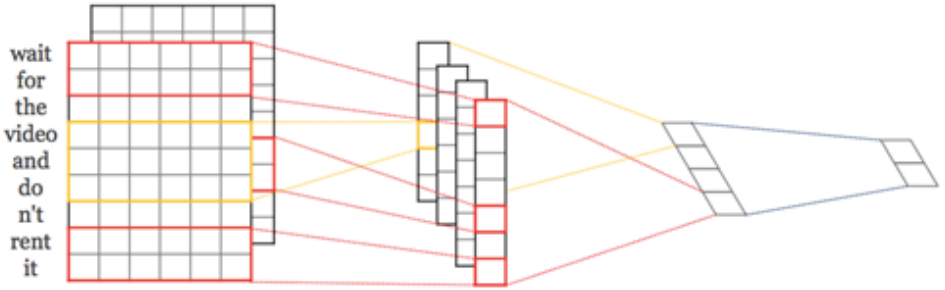
Summary of this lecture

Knowledge

I. Convolutional neural networks

A shallow CNN

A deep CNN

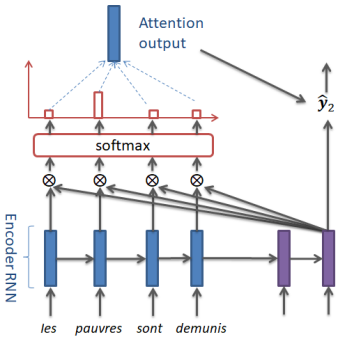


II. Recurrent neural networks

RNN for classification

Seq2seq for NMT

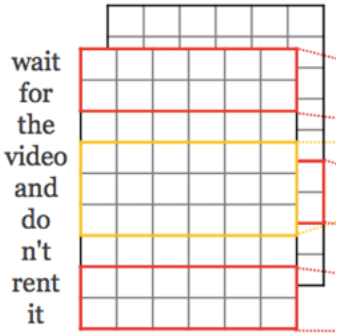
Seq2seq w/ attention



Summary of this lecture

Capability and value

- Comparative thinking





- Simple idea, great influence

TITLE	CITED BY	YEAR
Convolutional Neural Networks for Sentence Classification Y Kim EMNLP 2014	20556 *	2014

References

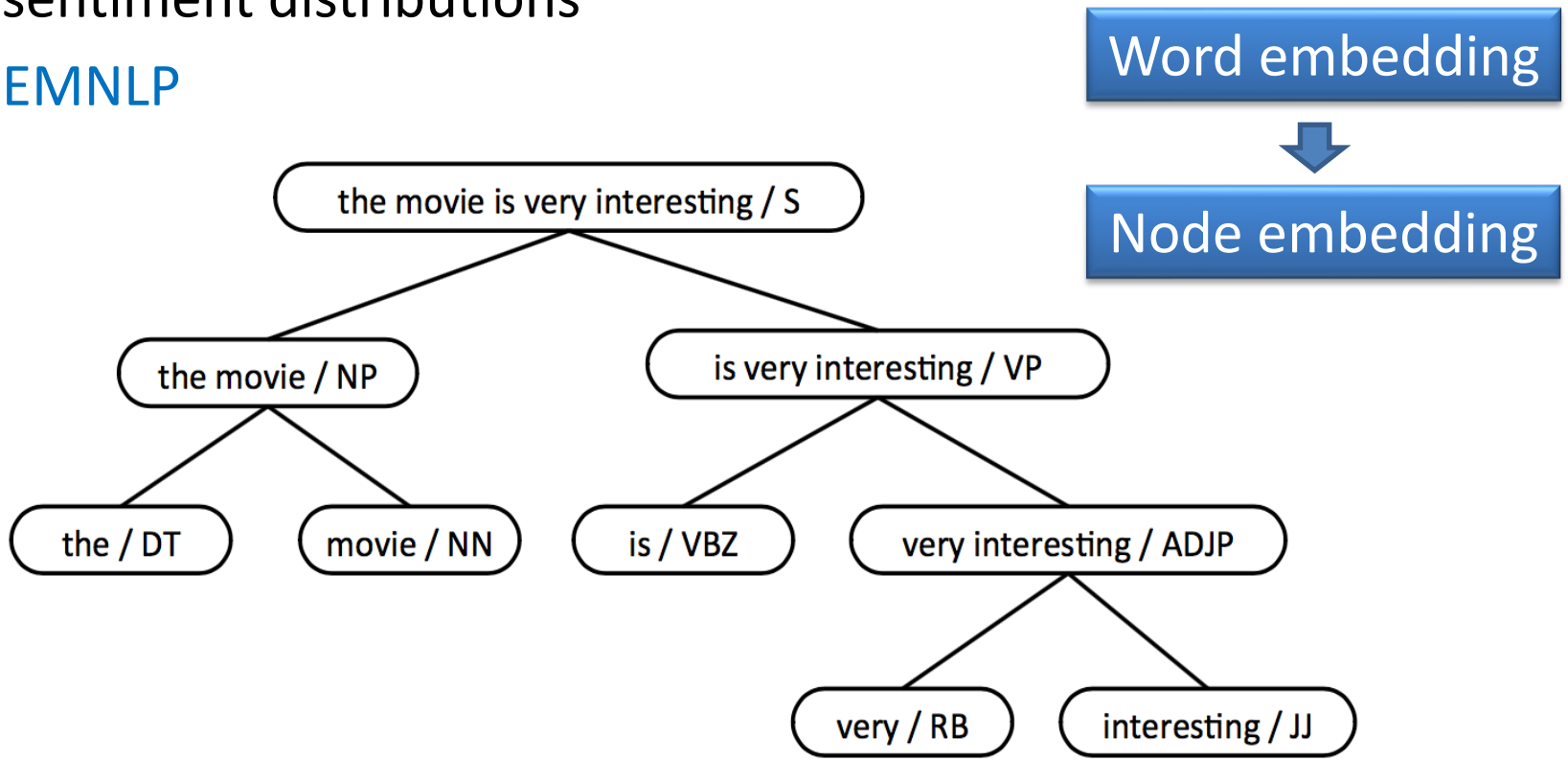
- Kim (2014)
  Convolutional neural networks for sentence classification
  EMNLP
- Kalchbrenner, Grefenstette, Blunsom (2014)
  A convolutional neural network for modelling sentences
  ACL
- Sutskever, Vinyals, Le (2015)
  Sequence to sequence learning with neural networks
  NeurIPS
- Bahdanau, Cho, Bengio (2015)
  Neural machine translation by jointly learning to align and translate
  ICLR

Recommended reading

- Socher, Pennington, Huang, Ng, Manning (2011)

  Semi-supervised recursive autoencoders for predicting sentiment distributions

  EMNLP



Homework 4

- RNN for sentence classification
  - Use Pytorch or Mindspore
  - Deadline: Dec 15 (Monday noon) 12:00